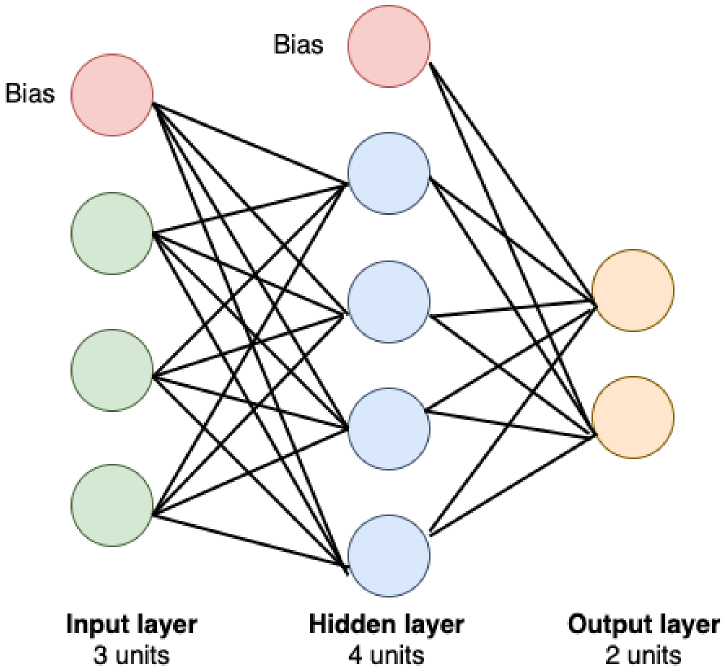
Hyperparameters & Parameters

Soft Computing

Parameters

- ❖ **Internal to the model and whose value can be estimated from data**
- ❖ **Required by the model when making predictions**
- ❖ **Learned from data**
- ❖ **Often not set manually by the practitioner**
- ❖ **Often saved as part of the learned model**
- ❖ **Define the skill of the model on your problem**
- ❖ **Examples:**
    - ➢ **Weight matrix of a model**
    - ➢ **Coefficients in a linear regression**
    - ➢ **Depth of a Decision tree**

Parameters


Bias
Bias
Input layer
3 units
Hidden layer
4 units
Output layer
2 units

Parameters

$$3 \times 4 + 4 \times 2 + 1 \times 4 + 1 \times 2$$

$$= 12 + 8 + 4 + 2$$

$$= 26$$

Hyperparameters

External to the model and whose value cannot be estimated from data

- ❖ **Set before training**
- ❖ **Determines the network structure**
    - ➢ **used in processes to help estimate model parameters**
- ❖ **Determine how the network is trained**
- ❖ **Tuned for a given predictive modeling problem**
- ❖ **Examples**
    - ➢ **Related to Network Structure**

■ **Dropout, # of Hidden Layer, Weight Initialization, Active Function, etc**

- ➢ **Related to Training**

■ **Epoch, Batch Size, Optimizer, Learning Rate**

Tuning Techniques

❖ **Random Search**

- **Randomly chooses combinations of hyperparameters from searchable space (all possible combinations).**

❖ **Grid Search**

- **Tests all possible combinations of hyperparameters of given Machine Learning algorithm**

Tuning Techniques

- ❖ **Bayesian optimizer**
  - **Based upon Bayes Rule and considers previously known knowledge to help narrow down the search space of good hyperparameter combinations.**
- ❖ **Gradient-based**
  - **A methodology to optimise several hyperparameters, based on the computation of the gradient of a machine learning model selection criterion with respect to the hyperparameters.**

Useful Links

https://machinelearningmastery.com/difference-between-a-parameter-and-a-hyper

parameter/

https://towardsdatascience.com/what-are-hyperparameters-and-how-to-tune-the-h

Yperparameters-in-a-deep-neural-network-d0604917584a

https://towardsdatascience.com/number-of-parameters-in-a-feed-forward-neural-network-4e4e33a53655